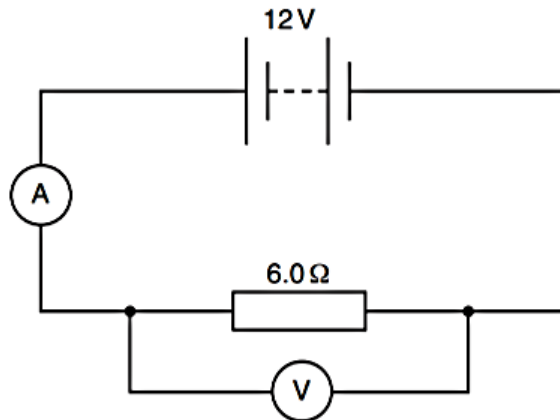


23 In the circuit shown, the battery is ideal but the ammeter and voltmeter are **not** ideal.



The ammeter reads 1.9 A while the voltmeter reads 11 V.

[Turn over

What are the resistances of the ammeter and voltmeter?				
		resistance of ammeter / Ω	resistance of voltmeter / Ω	
	A	0.32	170	
	B	0.32	8.3	
	C	0.53	170	
	D	0.53	8.3	

Ans: C

pd across ammeter = 1.0 V

resistance of ammeter = $1.0 \text{ V} \div 1.90 \text{ A} = 0.526 \Omega$

current across resistor = $11.0 \text{ V} \div 6.0 \Omega = 1.8333 \text{ A}$

current across voltmeter = $1.90 - 1.8333 = 0.0667 \text{ A}$

resistance of voltmeter = $11.0 \text{ V} \div 0.0667 \text{ A} = 165 \Omega$

alternatively, effective resistance of voltmeter and resistor = $11 \times$ resistance of ammeter